



Brief paper

The adaptive distributed observer approach to the cooperative output regulation of linear multi-agent systems[☆]He Cai^{a,b}, Frank L. Lewis^{c,d}, Guoqiang Hu^b, Jie Huang^a^a Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong, China^b School of Electrical and Electronic Engineering, Nanyang Technological University, 639798 Singapore, Singapore^c UTA Research Institute, University of Texas at Arlington, Fort Worth, Texas, USA^d State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang, China

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ABSTRACT

The cooperative output regulation problem of linear multi-agent systems using distributed observer approach has been solved under the assumption that each follower knows the system matrix of the leader system. To remove this assumption, in this paper, a distributed control scheme utilizing the adaptive distributed observer has been proposed, which consists of three steps. First, an adaptive distributed observer is designed to estimate both the system matrix and the state of the leader system. Second, an adaptive algorithm is developed to calculate the solutions of the regulator equations online. Third, both adaptive state feedback and adaptive measurement output feedback controllers are synthesized to solve the cooperative output regulation problem without the assumption that each follower knows the system matrix of the leader system.

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1. Introduction

The output regulation problem has been one of the central research topics for control systems (Davison, 1976; Francis, 1977; Francis & Wonham, 1976; Huang, 2004), which has numerous applications such as robot arm manipulation, satellite attitude control, and motor speed regulation. The cooperative output regulation problem, which can be viewed as an extension of the output regulation problem from a single plant to a multi-agent system, was further studied in Su and Huang (2012a,b) using the distributed observer approach. Specifically, the problem was first studied for the case of static communication network in Su and Huang (2012a), and then for the case of switching communication

network in Su and Huang (2012b). Other extensions of Su and Huang (2012a,b) can be found in Haghshenas, Badamchizadeh, and Baradarannia (2015), Li, Chen, and Ding (2015), Seyboth, Ren, and Allgower (2016). Specifically, the distributed observer was extended for multiple leaders in Haghshenas et al. (2015) to solve the containment control problem for linear multi-agent systems. In Li et al. (2015), the observer gain for the distributed observer is updated adaptively. In Seyboth et al. (2016), the agents are subject to both global and local external signals and distributed observers are adopted to estimate references and/or disturbances generated by heterogeneous exosystems.

The core of the distributed observer approach is the design of a distributed observer for the leader system of the following form,

$$\dot{v} = Sv \quad (1)$$

where $v \in R^q$ is the state of the leader system representing the reference input to be tracked and/or the external disturbance to be rejected and $S \in R^{q \times q}$ is a constant matrix. The distributed observer is capable of producing the estimation of the leader's signal to each follower so that a distributed control law can be synthesized to solve the problem. However, a drawback associated with the distributed observer approach is that each follower needs to know the matrix S (Haghshenas et al., 2015; Li et al., 2015; Seyboth et al., 2016; Su & Huang, 2012a,b), which might not be realistic in some applications. For example, if the leader's signal is

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a sinusoidal signal, then the assumption that each follower knows the matrix S amounts to assuming that the control of each follower knows the frequency of the sinusoidal signal. Thus, in this paper, we will further study the cooperative output regulation problem without requiring each follower knows the matrix S .

Our idea is to make use of some adaptive control technique to provide the estimation of the matrix S to each follower. To this end, we need to overcome three technical challenges. First, we employ the adaptive distributed observer proposed in [Cai and Huang \(2016\)](#) to provide the estimation of the matrix S . However, the adaptive distributed observer in [Cai and Huang \(2016\)](#) relies on two assumptions that all the eigenvalues of S are semi-simple with zero real parts and the communication graph of the followers is undirected. Clearly, these two assumptions are restrictive and our previous papers ([Su & Huang, 2012a,b](#)) do not rely on these two assumptions. Therefore, we will first remove these two assumptions by establishing [Lemmas 1 and 2](#). As a result, we are able to handle the frequently used signals such as polynomial signals including ramp signals. Second, since S is not available to each follower, the solution of the regulator equations, which determines the feedforward control gains for both the state feedback and measurement output feedback controllers, is not available either. We have to devise an algorithm to adaptively calculate the solution of the regulator equations based on the estimation of the matrix S . This result is summarized in [Lemmas 3 and 4](#). Finally, we need to synthesize both adaptive distributed state feedback and adaptive distributed measurement output feedback controllers utilizing the adaptive distributed observer and the estimated feedforward gain, and show that these controllers solve the cooperative output regulation problem without the assumption that each follower knows the matrix S .

It is noted that the design of the distributed observers in [Su and Huang \(2012a,b\)](#) is somehow related to the leader-following consensus problem studied in, say, [Ni and Cheng \(2010\)](#), [Zhang, Lewis, and Das \(2011\)](#) with $B = I$. However, the design of the adaptive distributed observer has not been encountered before since not all the followers know the matrix S .

The rest of this paper is organized as follows. Problem formulation is introduced in Section 2. Section 3 presents the design of the adaptive distributed observer. Section 4 shows how to solve the cooperative output regulation problem using the adaptive distributed observer. A numerical example is given in Section 5 to validate the effectiveness of the proposed control laws. Section 6 concludes this paper.

2. Problem formulation

2.1. Notation

$\otimes$ denotes the Kronecker product of matrices. 1_N denotes an N dimensional column vector whose components are all 1. $\|x\|$ denotes the Euclidean norm of a vector x and $\|A\|$ denotes the Euclidean norm of a matrix A . For $x_i \in \mathbb{R}^{n_i \times p}$, $i = 1, \dots, m$, $\text{col}(x_1, \dots, x_m) = [x_1^T, \dots, x_m^T]^T$. $\sigma(A)$ denotes the spectrum of a square matrix A . $\Re(\sigma(A))$ denotes the set of the real parts of all the elements of $\sigma(A)$. For any matrix $A \in \mathbb{R}^{m \times n}$, $\text{vec}(A) = \text{col}(A_1, \dots, A_n)$ where $A_i \in \mathbb{R}^m$ is the i th column of A . For any column vector $X \in \mathbb{R}^{nq}$ for some positive integers n and q , $M_n^q(X) = [X_1, \dots, X_q]$, where, for $i = 1, \dots, q$, $X_i \in \mathbb{R}^n$, and are such that $X = \text{col}(X_1, \dots, X_q)$. For $i = 1, 2$, let $A_i(t) : [0, \infty) \rightarrow \mathbb{R}^{n_i \times p_i}$ be any two piecewise continuous matrices such that, for some positive numbers α_i and β_i , $\|A_i(t)\| \leq \beta_i e^{-\alpha_i t}$ for all $t \geq 0$. Then α_i is called the rate of exponential convergence of $A_i(t)$, and $A_1(t)$ is said to decay to zero exponentially at least as fast as $A_2(t)$ if $\alpha_1 \geq \alpha_2$.

A digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a finite set of nodes $\mathcal{V} = \{1, \dots, N\}$ and an edge set $\mathcal{E} = \{(i, j), i, j \in \mathcal{V}, i \neq j\}$. An edge from node i to node j is denoted by (i, j) , and node i is called the neighbor of node j . If the digraph $\mathcal{G}$ contains a sequence of edges of the form $(i_1, i_2), (i_2, i_3), \dots, (i_k, i_{k+1})$, then the set $\{(i_1, i_2), (i_2, i_3), \dots, (i_k, i_{k+1})\}$ is called a directed path of $\mathcal{G}$ from i_1 to i_{k+1} , and node i_{k+1} is said to be reachable from node i_1 . A digraph is said to contain a spanning tree if there exists a node i such that any other node is reachable from node i . The node i is called the root of the spanning tree. The edge (i, j) is called undirected if $(i, j) \in \mathcal{E} \Leftrightarrow (j, i) \in \mathcal{E}$. The graph is called undirected if every edge in $\mathcal{E}$ is undirected. A digraph $\mathcal{G}_s = (\mathcal{V}_s, \mathcal{E}_s)$ is called a subgraph of $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ if $\mathcal{V}_s \subseteq \mathcal{V}$ and $\mathcal{E}_s \subseteq \mathcal{E} \cap (\mathcal{V}_s \times \mathcal{V}_s)$. The weighted adjacency matrix $\mathcal{A}^w = [a_{ij}] \in \mathbb{R}^{N \times N}$ of $\mathcal{G}$ is defined as $a_{ii} = 0$; for $i \neq j$, $a_{ij} > 0 \Leftrightarrow (j, i) \in \mathcal{E}$ and $a_{ij} = a_{ji}$ if (i, j) is an undirected edge of $\mathcal{E}$. The Laplacian of $\mathcal{G}$ is defined as $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{N \times N}$, where $l_{ii} = \sum_{j=1}^N a_{ij}$, $l_{ij} = -a_{ij}$ for $i \neq j$.

2.2. Problem formulation

As in [Su and Huang \(2012a\)](#), we consider the following linear heterogeneous multi-agent system:

$$\dot{x}_i = A_i x_i + B_i u_i + E_i v \quad (2a)$$

$$e_i = C_i x_i + D_i u_i + F_i v \quad (2b)$$

$$y_{mi} = C_{mi} x_i + D_{mi} u_i + F_{mi} v, \quad i = 1, \dots, N \quad (2c)$$

where $x_i \in \mathbb{R}^{n_i}$, $u_i \in \mathbb{R}^{m_i}$, $e_i \in \mathbb{R}^{p_i}$, $y_{mi} \in \mathbb{R}^{p_{mi}}$ are the state, control input, regulated output and measurement output of the i th subsystem, respectively.

Also as in [Su and Huang \(2012a\)](#), we treat the system composed of (1) and (2) as a multi-agent system of $(N + 1)$ agents with (1) as the leader and the N subsystems of (2) as N followers, and define a graph $\bar{\mathcal{G}} = (\bar{\mathcal{V}}, \bar{\mathcal{E}})$ with $\bar{\mathcal{V}} = \{0, 1, \dots, N\}$ and $\bar{\mathcal{E}} \subseteq \bar{\mathcal{V}} \times \bar{\mathcal{V}}$. Here the node 0 is associated with the leader system (1) and the node i , $i = 1, \dots, N$, is associated with the i th subsystem of the follower system (2). For $i = 0, 1, \dots, N$, $j = 1, \dots, N$, $(i, j) \in \bar{\mathcal{E}}$ if and only if agent j can use the information of agent i for control. We can further define a subgraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of $\bar{\mathcal{G}}$ where $\mathcal{V} = \{1, \dots, N\}$ and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is obtained from $\bar{\mathcal{E}}$ by removing all the edges between node 0 and the nodes in $\mathcal{V}$. In terms of $\bar{\mathcal{G}}$, we can define the following cooperative output regulation problem.

Problem 1. Given systems (1), (2) and a graph $\bar{\mathcal{G}}$, design a distributed control law u_i , such that the trajectory of the closed-loop system starting from any initial condition exists for all $t \geq 0$ and satisfies, (i) when v is bounded, the trajectory of the closed-loop system is bounded for all $t \geq 0$; (ii) the regulated output satisfies $\lim_{t \rightarrow \infty} e_i(t) = 0$ for $i = 1, \dots, N$.

We need the following assumptions for the solvability of [Problem 1](#).

Assumption 1. (A_i, B_i) are stabilizable, $i = 1, \dots, N$.

Assumption 2. (C_{mi}, A_i) are detectable, $i = 1, \dots, N$.

Assumption 3. The following linear matrix equations

$$X_i S = A_i X_i + B_i U_i + E_i \quad (3a)$$

$$0 = C_i X_i + D_i U_i + F_i \quad (3b)$$

have solution pairs (X_i, U_i) , respectively.

Assumption 4. The graph $\bar{\mathcal{G}}$ contains a spanning tree with the node 0 as the root.

Remark 1. In the classical linear output regulation problem, Eqs. (3) are called the regulator equations whose solvability imposes a necessary condition for the solvability of the output regulation problem (Davison, 1976; Francis, 1977; Francis & Wonham, 1976). The solution pair (X_i, U_i) determines the feedforward control gains for both the state feedback and measurement output feedback controllers. However, in our case, since not all the followers can access the system matrix S , we cannot calculate (X_i, U_i) , and hence the feedforward gains in advance. To overcome this difficulty, we will develop an adaptive algorithm to solve (3) online.

3. Adaptive distributed observer

The distributed observer for the leader system was first developed in Su and Huang (2012a) for the static network topology and then in Su and Huang (2012b) for the switching network topology. Let $\bar{A}^w = [a_{ij}]$ denote the weighted adjacency matrix of $\bar{\mathcal{G}}$. Then the distributed observer for (1) proposed in Su and Huang (2012a) is as follows

$$\dot{\eta}_i = S\eta_i + \mu \sum_{j=0}^N a_{ij}(\eta_j - \eta_i), \quad i = 1, \dots, N \quad (4)$$

where $\eta_0 = v$, $\eta_i \in \mathbb{R}^q$, $i = 1, \dots, N$, and $\mu > 0$. By Theorem 1 of Su and Huang (2012a), under Assumption 4, for sufficiently large μ , we have $\lim_{t \rightarrow \infty} (\eta_i(t) - v(t)) = 0$, $i = 1, \dots, N$. That is why (4) is called the distributed observer for (1).

However, in (4), the matrix S is used by every follower, which may not be realistic in some applications. In Cai and Huang (2016), an adaptive distributed observer was further proposed for switching network topology. A special case of the adaptive distributed observer in Cai and Huang (2016) for static network topology takes the following form

$$\dot{\hat{S}}_i = \mu_1 \sum_{j=0}^N a_{ij}(\hat{S}_j - \hat{S}_i) \quad (5a)$$

$$\dot{\eta}_i = \hat{S}_i \eta_i + \mu_2 \sum_{j=0}^N a_{ij}(\eta_j - \eta_i) \quad (5b)$$

where $S_0 = S$, $\hat{S}_i \in \mathbb{R}^{q \times q}$, $i = 1, \dots, N$, $\mu_1, \mu_2 > 0$. Under Assumption 4, and the assumptions that all the eigenvalues of S are semi-simple with zero real parts and $\mathcal{G}$ is undirected, it was shown that, for $i = 1, \dots, N$, $\lim_{t \rightarrow \infty} (\hat{S}_i(t) - S) = 0$, $\lim_{t \rightarrow \infty} (\eta_i(t) - v(t)) = 0$. It is noted that, this observer incorporates a mechanism for estimating the matrix S , and is thus called the adaptive distributed observer. From (5a), it can be seen that only those followers who are the children of the leader know the matrix S . Thus, the adaptive distributed observer is more realistic than the distributed observer.

In this section, we will remove the assumption that all the eigenvalues of S are semi-simple with zero real parts so that we can also handle unbounded signals such as ramp signals. Moreover, we will also remove the assumption that $\mathcal{G}$ is undirected. Let us first establish the following lemma.

Lemma 1. Consider the following system

$$\dot{x} = \epsilon Fx + F_1(t)x + F_2(t) \quad (6)$$

where $x \in \mathbb{R}^n$, $F \in \mathbb{R}^{n \times n}$ is Hurwitz, $\epsilon > 0$, $F_1(t) \in \mathbb{R}^{n \times n}$ and $F_2(t) \in \mathbb{R}^n$ are bounded and continuous for all $t \geq t_0$. We have (i) if $F_1(t), F_2(t) \rightarrow 0$ as $t \rightarrow \infty$ (exponentially), then, for any $x(t_0)$ and any $\epsilon > 0$, $x(t) \rightarrow 0$ as $t \rightarrow \infty$ (exponentially); (ii) if $F_1(t) = 0$, $F_2(t)$ decays to zero exponentially at the rate of α , and $\epsilon > \frac{\alpha}{\alpha_F}$, where $\alpha_F = \min(\Re(\sigma(-F)))$, then, for any $x(t_0)$, $x(t) \rightarrow 0$ as $t \rightarrow \infty$ exponentially at the rate of α .

Proof. Part (i). Consider the following system

$$\dot{x} = \epsilon Fx + F_1(t)x \triangleq F_3(t)x. \quad (7)$$

Since ϵF is Hurwitz, the origin of the system $\dot{x} = \epsilon Fx$ is exponentially stable. Let $\bar{P} \in \mathbb{R}^{n \times n}$ be the positive definite solution of $\epsilon \bar{P}F + \epsilon F^T \bar{P} = -I_n$ and $V_P = x^T \bar{P}x$. Then $\min(\sigma(\bar{P}))\|x\|^2 \leq V_P \leq \max(\sigma(\bar{P}))\|x\|^2$, $\frac{\partial V_P}{\partial x} \epsilon Fx \leq -\frac{1}{2}\|x\|^2$ and $\|\frac{\partial V_P}{\partial x}\| \leq \|\bar{P}\| \cdot \|x\|$. Note that $\|F_1(t)x\| \leq \|F_1(t)\| \cdot \|x\|$. Since $F_1(t) \rightarrow 0$ as $t \rightarrow \infty$, by case 2 of Lemma 9.5 and Corollary 9.1 of Khalil (2002), the origin of (7) is globally exponentially stable. Therefore, by Theorem 4.12 of Khalil (2002), there exist positive definite matrices $\bar{Q}_1(t), \bar{Q}_2(t) \in \mathbb{R}^{n \times n}$ satisfying $0 < \underline{q}_1 I_n \leq \bar{Q}_1(t) \leq \bar{q}_1 I_n$, $0 < \underline{q}_2 I_n \leq \bar{Q}_2(t) \leq \bar{q}_2 I_n$ for some $\underline{q}_1, \bar{q}_1, \underline{q}_2, \bar{q}_2 > 0$ and for all $t \geq t_0$, such that $\dot{\bar{Q}}_1(t) + \bar{Q}_1(t)F_3(t) + F_3(t)^T \bar{Q}_1(t) = -\bar{Q}_2(t)$. Let $V_Q = x^T \bar{Q}_1(t)x$. Then along the trajectory of (6), we have

$$\begin{aligned} \dot{V}_Q &= -x^T \bar{Q}_2(t)x + 2x^T \bar{Q}_1(t)F_2(t) \\ &\leq -\underline{q}_2 \|x\|^2 + \frac{\underline{q}_2}{2} \|x\|^2 + \frac{2\bar{q}_1^2}{\underline{q}_2} \|F_2(t)\|^2 \\ &= -\frac{\underline{q}_2}{2} \|x\|^2 + \frac{2\bar{q}_1^2}{\underline{q}_2} \|F_2(t)\|^2. \end{aligned} \quad (8)$$

Therefore, by Theorem 2.7 of Chen and Huang (2015), system (6) is input-to-state stable with $F_2(t)$ as the input. Since $F_2(t)$ is bounded, the solution of (6) is bounded for any initial condition. Since the origin of $\dot{x} = \epsilon Fx$ is also exponentially stable, system (6) is input-to-state stable with $F_4(t) \triangleq F_1(t)x + F_2(t)$ as the input. Thus it has asymptotic gain property (Definition 2.11 of Chen & Huang, 2015), which implies the existence of a class $\mathcal{K}$ function κ such that, for any $x(t_0) \in \mathbb{R}^n$, $x(t)$ satisfies

$$\lim_{t \rightarrow \infty} \left[\sup_{t \geq \tau} \|x(t)\| \right] \leq \kappa \left[\lim_{\tau \rightarrow \infty} \left(\sup_{t \geq \tau} \|F_4(t)\| \right) \right]. \quad (9)$$

Since $x(t)$ is bounded, if $F_1(t), F_2(t)$ tend to zero (exponentially), so does $x(t)$.

Part (ii). The solution of (6) with $F_1(t) = 0$ is given by $x(t) = x_t(t) + x_s(t)$ where $x_t(t) = e^{\epsilon F t} x(0)$ and $x_s(t) = \int_0^t e^{\epsilon F(t-\tau)} F_2(\tau) d\tau$. Since $\epsilon \alpha_F > \alpha$, $x_t(t)$ decays to zero exponentially at the rate of α . Moreover, $\|x_s(t)\| \leq \int_0^t \beta_s \beta e^{-\epsilon \alpha_F(t-\tau)} \beta e^{-\alpha \tau} d\tau$ for some $\beta_s > 0$. Simple calculation gives $\|x_s(t)\| < \frac{\beta_s \beta}{\epsilon \alpha_F - \alpha} e^{-\alpha t}$ and hence the proof is completed. $\square$

Remark 2. Alternatively, the exponential stability of the origin of (7) can be directly obtained by Example 9.6 of Khalil (2002) and the input-to-state stability of the origin of (6) can be obtained by Examples 2.11 and 2.12 of Chen and Huang (2015). From the proof of Lemma 1, we can see that if system (7) is exponentially stable and $F_2(t)$ is bounded, then the solution of system (6) is bounded.

We now establish the main result of this section.

Lemma 2. Given the systems (1) and (5), let $\tilde{S}_i = S_i - S$, $\tilde{\eta}_i = \eta_i - v$. Then, for any initial condition $\tilde{S}_i(0)$ and $\tilde{\eta}_i(0)$, we have, (i) for any $\mu_1 > 0$, for $i = 1, \dots, N$, $\lim_{t \rightarrow \infty} \tilde{S}_i(t) = 0$ exponentially, and (ii) for sufficiently large μ_1 and $\mu_2 > 0$, for $i = 1, \dots, N$, $\lim_{t \rightarrow \infty} \tilde{\eta}_i(t) = 0$ exponentially.

Proof. Part (i). Let $\mathcal{L}$ be the Laplacian of $\mathcal{G}$, $H = \mathcal{L} + \text{diag}\{a_{10}, \dots, a_{N0}\}$, and $\tilde{S} = \text{col}(\tilde{S}_1, \dots, \tilde{S}_N)$. Then (5a) can be put in the form of $\dot{\tilde{S}} = -\mu_1(H \otimes I_q)\tilde{S}$ or a more standard form $\text{vec}(\dot{\tilde{S}}) = -\mu_1(I_q \otimes H \otimes I_q)\text{vec}(\tilde{S})$. Under Assumption 4, by Lemma 4 of Hu and Hong (2007), all the eigenvalues of H have positive real parts. Therefore, $\lim_{t \rightarrow \infty} \text{vec}(\tilde{S}(t)) = 0$ exponentially, which implies, for

$i = 1, \dots, N$, $\lim_{t \rightarrow \infty} \tilde{S}_i(t) = 0$ exponentially. Moreover, let $\alpha_H = \min(\Re(\sigma(H)))$. Then, for all $t \geq 0$, $\|\tilde{S}_i(t)\| \leq \beta_H \|\tilde{S}_i(0)\| e^{-\mu_1 \alpha_H t}$ for some $\beta_H > 0$.

Part (ii). By (5), we have

$$\begin{aligned} \dot{\tilde{\eta}}_i &= S_i \eta_i - S v + \mu_2 \sum_{j=0}^N a_{ij}(\tilde{\eta}_j - \tilde{\eta}_i) \\ &= S_i \eta_i - S \eta_i + S \eta_i - S v + \mu_2 \sum_{j=0}^N a_{ij}(\tilde{\eta}_j - \tilde{\eta}_i) \\ &= S \tilde{\eta}_i + \tilde{S}_i \eta_i + \mu_2 \sum_{j=0}^N a_{ij}(\tilde{\eta}_j - \tilde{\eta}_i) \\ &= S \tilde{\eta}_i + \tilde{S}_i \tilde{\eta}_i + \tilde{S}_i v + \mu_2 \sum_{j=0}^N a_{ij}(\tilde{\eta}_j - \tilde{\eta}_i). \end{aligned} \quad (10)$$

Let $\tilde{\eta} = \text{col}(\tilde{\eta}_1, \dots, \tilde{\eta}_N)$ and $\tilde{S}_d = \text{block diag}\{\tilde{S}_1, \dots, \tilde{S}_N\}$. Then (10) can be rewritten into the following compact form

$$\dot{\tilde{\eta}} = [(I_N \otimes S) - \mu_2(H \otimes I_q)]\tilde{\eta} + \tilde{S}_d \tilde{\eta} + \tilde{S}_d(1_N \otimes v). \quad (11)$$

If none of the eigenvalues of S has positive real parts, then, for any positive μ_1 and μ_2 , the matrix $(I_N \otimes S) - \mu_2(H \otimes I_q)$ is Hurwitz (Su & Huang, 2012a) and $\tilde{S}_d(t)(1_N \otimes v(t))$ will decay to zero exponentially. If some of the eigenvalues of S have positive real parts, let $\gamma = \max(\Re(\sigma(S)))$. Then, for $\mu_2 > \frac{\gamma}{\alpha_H}$ and $\mu_1 > \frac{\gamma}{\alpha_H}$, the matrix $(I_N \otimes S) - \mu_2(H \otimes I_q)$ is still Hurwitz and $\tilde{S}_d(t)(1_N \otimes v(t))$ will still decay to zero exponentially. Thus, by Lemma 1, $\lim_{t \rightarrow \infty} \tilde{\eta}_i(t) = 0$ exponentially and the proof is completed. $\square$

4. Main result

4.1. Adaptive Solution of regulator equations

Our control law needs to use the solution of the regulator equations to provide an appropriate feedforward control to achieve the control objective. The solution of the regulator equations relies on the system matrix S of the leader system. Since those followers which are not the children of the leader do not know S , we propose to use S_i , the estimation of S , to adaptively calculate the solution of the regulator equations. For this purpose, we first introduce the following lemma.

Lemma 3. Let $A \in R^{m \times n}$, $b \in R^m$, and $\text{rank}(A) = \text{rank}(A, b) = k$ for some positive integers $m, n, k \geq 1$, and let $\mathcal{A}(t) \in R^{m \times n}$ be any bounded and piecewise continuous time-varying matrix such that $\tilde{A}(t) \triangleq (\mathcal{A}(t) - A) \rightarrow 0$ as $t \rightarrow \infty$ exponentially at the rate of α . Then, for any $x(t_0) \in R^n$ and $\varepsilon > 0$, the following system

$$\dot{x} = -\varepsilon \mathcal{A}(t)^T (\mathcal{A}(t)x - b) \quad (12)$$

has a unique bounded solution $x(t)$ for $t \geq t_0$ such that, for some $x^* \in R^n$ satisfying $Ax^* = b$, $\lim_{t \rightarrow \infty} (x(t) - x^*) = 0$ exponentially. In particular, if ε is sufficiently large, then $x(t) - x^* \rightarrow 0$ as $t \rightarrow \infty$ exponentially at the rate of α .

Proof. By singular value decomposition method, there exists an orthogonal matrix $P \in R^{n \times n}$ such that $AP = [\bar{A} \ 0_{m \times (n-k)}]$ where $\bar{A} \in R^{m \times k}$. Then we have

$$P^T A^T A P = \begin{bmatrix} \bar{A}^T \bar{A} & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & 0_{(n-k) \times (n-k)} \end{bmatrix} \quad (13)$$

and

$$P^T A^T b = \begin{bmatrix} \bar{A}^T b \\ 0_{(n-k) \times 1} \end{bmatrix}. \quad (14)$$

Since $\bar{A}$ has full column rank, there exists a unique $\bar{x}_1^* \in R^k$ such that $\bar{A}\bar{x}_1^* = b$. Let $\bar{x}_2^*$ be any column vector of dimension $(n-k)$, and $x^* = P \begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix}$. Then

$$Ax^* = APP^T x^* = [\bar{A} \ 0] \begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix} = b. \quad (15)$$

Next, let $\bar{x} = P^T x$. Then,

$$\begin{aligned} \dot{\bar{x}} &= P^T (-\varepsilon \mathcal{A}(t)^T \mathcal{A}(t)x + \varepsilon \mathcal{A}(t)^T b) \\ &= -\varepsilon P^T \mathcal{A}(t)^T \mathcal{A}(t)x + \varepsilon P^T \mathcal{A}(t)^T b \\ &= -\varepsilon P^T A^T A x + \varepsilon P^T A^T A x - \varepsilon P^T \mathcal{A}(t)^T \mathcal{A}(t)x \\ &\quad + \varepsilon P^T \mathcal{A}(t)^T b - \varepsilon P^T A^T b + \varepsilon P^T A^T b \\ &= -\varepsilon P^T A^T A P \bar{x} + \varepsilon P^T (A^T A - \mathcal{A}(t)^T \mathcal{A}(t))x \\ &\quad + \varepsilon P^T \tilde{A}(t)^T b + \varepsilon P^T A^T b \\ &= -\varepsilon P^T A^T A P \bar{x} + \varepsilon P^T A^T b + d(t) \end{aligned} \quad (16)$$

where $d(t) = \varepsilon P^T (A^T A - \mathcal{A}(t)^T \mathcal{A}(t))P \bar{x} + \varepsilon P^T \tilde{A}(t)^T b$. Then, $\bar{x}$ is bounded by Remark 2, and thus, $\lim_{t \rightarrow \infty} d(t) = 0$ exponentially at the rate of α . Next, let $\bar{x} = \text{col}(\bar{x}_1, \bar{x}_2)$ and $d(t) = \text{col}(d_1(t), d_2(t))$ with $\bar{x}_1, d_1(t) \in R^k$ and $\bar{x}_2, d_2(t) \in R^{n-k}$. Then by (13) and (14), (16) can be rewritten as

$$\dot{\bar{x}}_1 = -\varepsilon \bar{A}^T \bar{A} \bar{x}_1 + \varepsilon \bar{A}^T b + d_1(t) \quad (17a)$$

$$\dot{\bar{x}}_2 = d_2(t). \quad (17b)$$

Clearly, there exists $\bar{x}_2^* \in R^{n-k}$ such that $\lim_{t \rightarrow \infty} (\bar{x}_2(t) - \bar{x}_2^*) = 0$ exponentially at the rate of α . Let $\tilde{x}_1 = \bar{x}_1 - \bar{x}_1^*$. Then we have

$$\begin{aligned} \dot{\tilde{x}}_1 &= -\varepsilon \bar{A}^T \bar{A} \tilde{x}_1 + \varepsilon \bar{A}^T b + d_1(t) \\ &= -\varepsilon \bar{A}^T \bar{A} \tilde{x}_1 - \varepsilon \bar{A}^T \bar{A} \bar{x}_1^* + \varepsilon \bar{A}^T b + d_1(t) \\ &= -\varepsilon \bar{A}^T \bar{A} \tilde{x}_1 - \varepsilon \bar{A}^T (\bar{A} \bar{x}_1^* - b) + d_1(t) \\ &= -\varepsilon \bar{A}^T \bar{A} \tilde{x}_1 + d_1(t). \end{aligned} \quad (18)$$

Since $-\bar{A}^T \bar{A}$ is Hurwitz, and $\lim_{t \rightarrow \infty} d_1(t) = 0$ exponentially at the rate of α , by Lemma 1, for any $\varepsilon > 0$, $\lim_{t \rightarrow \infty} \tilde{x}_1(t) = 0$ exponentially. In particular, by Lemma 1 again, with $\varepsilon > \frac{\alpha}{\min(\sigma(\bar{A}^T \bar{A}))}$, $\lim_{t \rightarrow \infty} \tilde{x}_1(t) = 0$ exponentially at the rate of α .

Let $x^* = P \begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix}$. Then $Ax^* = b$ from (15) and

$$\begin{aligned} \lim_{t \rightarrow \infty} (x(t) - x^*) &= \lim_{t \rightarrow \infty} \left(P \bar{x}(t) - P \begin{bmatrix} \bar{x}_1^* \\ \bar{x}_2^* \end{bmatrix} \right) \\ &= \lim_{t \rightarrow \infty} P \begin{bmatrix} \bar{x}_1(t) - \bar{x}_1^* \\ \bar{x}_2(t) - \bar{x}_2^* \end{bmatrix} = 0 \end{aligned} \quad (19)$$

exponentially at least as fast as $\tilde{A}(t)$. $\square$

By letting $\mathcal{A}(t) = A$ for all $t \geq t_0$ in Lemma 3, we have the following result.

Corollary 1. Let $A \in R^{m \times n}$, $b \in R^m$, and $\text{rank}(A) = \text{rank}(A, b) = k$ for some positive integers $m, n, k \geq 1$. Then, for any $x(0) \in R^n$ and $\varepsilon > 0$, the following system

$$\dot{x} = -\varepsilon A^T (Ax - b) \quad (20)$$

has a unique bounded solution $x(t)$ for $t \geq 0$ such that, for some $x^* \in R^n$ satisfying $Ax^* = b$, $\lim_{t \rightarrow \infty} (x(t) - x^*) = 0$ exponentially.

Remark 3. Using a differential equation of the form (20) to minimize $\|Ax - b\|$ where $m \geq n$ is given in Cichocki and Unbehauen (1992). What makes Lemma 3 interesting is that it leads to a solution of the equation $Ax = b$ without using the exact knowledge of the matrix A and without assuming A is of full rank.

Next, we will show how to solve the regulator equations adaptively. For this purpose, for $i = 1, \dots, N$, let $\chi_i = \text{vec} \begin{pmatrix} X_i \\ U_i \end{pmatrix}$, $b_i = \text{vec} \begin{pmatrix} E_i \\ F_i \end{pmatrix}$ and

$$\mathcal{Q}_i(t) = S_i(t)^T \otimes \begin{bmatrix} I_{n_i} & 0 \\ 0 & 0 \end{bmatrix} - I_q \otimes \begin{bmatrix} A_i & B_i \\ C_i & D_i \end{bmatrix} \quad (21)$$

where S_i is provided by (5a). It is noted that the dimensions of χ_i and $\mathcal{Q}_i(t)$ are $q(n_i + m_i) \times 1$ and $q(n_i + p_i) \times q(n_i + m_i)$, respectively. We have the following result.

Lemma 4. Under Assumption 3, for $i = 1, \dots, N$, for any initial condition $\zeta_i(t_0)$, each of the following equations

$$\dot{\zeta}_i = -\mu_3 \mathcal{Q}_i(t)^T (\mathcal{Q}_i(t) \zeta_i - b_i) \quad (22)$$

where $\mu_3 > 0$ is sufficiently large, has a unique bounded solution over $t \geq t_0$. Moreover, let $\Xi_i(t) = M_{(n_i+m_i)}^q(\zeta_i(t))$. Then, for some solution (X_i^*, U_i^*) of the regulator equations (3), we have

$$\lim_{t \rightarrow \infty} \left(\Xi_i(t) - \begin{bmatrix} X_i^* \\ U_i^* \end{bmatrix} \right) = 0 \quad (23)$$

exponentially at least as fast as $\tilde{S}(t)$.

Proof. The regulator equations (3) can be put into the following form

$$\begin{bmatrix} I_{n_i} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X_i \\ U_i \end{bmatrix} S - \begin{bmatrix} A_i & B_i \\ C_i & D_i \end{bmatrix} \begin{bmatrix} X_i \\ U_i \end{bmatrix} = \begin{bmatrix} E_i \\ F_i \end{bmatrix}. \quad (24)$$

By Theorem 1.9 of Huang (2004), (24) can be transformed into the following form

$$\mathcal{Q}_i \chi_i = b_i \quad (25)$$

where

$$\mathcal{Q}_i = S^T \otimes \begin{bmatrix} I_{n_i} & 0 \\ 0 & 0 \end{bmatrix} - I_q \otimes \begin{bmatrix} A_i & B_i \\ C_i & D_i \end{bmatrix}. \quad (26)$$

Under Assumption 3, for $i = 1, \dots, N$, the linear algebraic equations (25) are solvable. From (21) and (26), $\lim_{t \rightarrow \infty} (\mathcal{Q}_i(t) - \mathcal{Q}_i) = 0$ exponentially at least as fast as $\tilde{S}(t)$. Therefore, by Lemma 3, for any $\zeta_i(t_0)$, (22) has a unique bounded solution $\zeta_i(t)$ for $t \geq t_0$ such that, for some χ_i^* satisfying (25), $\lim_{t \rightarrow \infty} (\zeta_i(t) - \chi_i^*) = 0$ exponentially at least as fast as $\tilde{S}(t)$. The proof is completed by noting

$$\left(\Xi_i(t) - \begin{bmatrix} X_i^* \\ U_i^* \end{bmatrix} \right) = M_{(n_i+m_i)}^q(\zeta_i(t) - \chi_i^*). \quad \square \quad (27)$$

4.2. Control law design and stability analysis

In this subsection, we will first show how to solve Problem 1 by state feedback control. Let $\Xi_i(t) = [\mathcal{X}_i(t)^T, \mathcal{U}_i(t)^T]^T$, where $\mathcal{X}_i(t) \in \mathbb{R}^{n_i \times q}$ and $\mathcal{U}_i(t) \in \mathbb{R}^{m_i \times q}$. Since (A_i, B_i) is stabilizable, let K_{xi}

be such that $A_i + B_i K_{xi}$ is Hurwitz and $K_{\eta i}(t)$ be given as $K_{\eta i}(t) = \mathcal{U}_i(t) - K_{xi} \mathcal{X}_i(t)$. For $i = 1, \dots, N$, we design the following state feedback controller

$$u_i = K_{xi} \mathcal{X}_i + K_{\eta i}(t) \eta_i. \quad (28)$$

We have the following result.

Theorem 1. Given systems (1), (2) and a graph $\bar{\mathcal{G}}$, under Assumptions 1, 3 and 4, for sufficiently large μ_1, μ_2 and μ_3 , Problem 1 is solvable by the control law composed of (5), (22), and (28).

Proof. Let $\tilde{x}_i = x_i - X_i^* v$ and $\tilde{u}_i = u_i - U_i^* v$, $K_{\eta i}^* = U_i^* - K_{xi} X_i^*$, and $\tilde{K}_{\eta i}(t) = K_{\eta i}(t) - K_{\eta i}^*$. By making use of the solution of the regulator equations (3), we obtain, for $i = 1, \dots, N$,

$$\begin{aligned} \dot{\tilde{x}}_i &= A_i \tilde{x}_i + B_i \tilde{u}_i + E_i v - X_i^* S v \\ &= A_i (\tilde{x}_i + X_i^* v) + B_i (\tilde{u}_i + U_i^* v) + E_i v - X_i^* S v \\ &= A_i \tilde{x}_i + B_i \tilde{u}_i \end{aligned} \quad (29)$$

and

$$\begin{aligned} e_i &= C_i \tilde{x}_i + D_i \tilde{u}_i + F_i v \\ &= C_i (\tilde{x}_i + X_i^* v) + D_i (\tilde{u}_i + U_i^* v) + F_i v \\ &= C_i \tilde{x}_i + D_i \tilde{u}_i. \end{aligned} \quad (30)$$

Further, by (28), we have

$$\begin{aligned} \tilde{u}_i &= K_{xi} (\tilde{x}_i + X_i^* v) + K_{\eta i}(t) (\tilde{\eta}_i + v) - U_i^* v \\ &= K_{xi} \tilde{x}_i + (U_i^* - K_{\eta i}^*) v + K_{\eta i}(t) (\tilde{\eta}_i + v) - U_i^* v \\ &= K_{xi} \tilde{x}_i + K_{\eta i}(t) \tilde{\eta}_i + \tilde{K}_{\eta i}(t) v. \end{aligned} \quad (31)$$

Substituting (31) into (29) gives

$$\dot{\tilde{x}}_i = (A_i + B_i K_{xi}) \tilde{x}_i + f_i(t) \quad (32)$$

where $f_i(t) = B_i K_{\eta i}(t) \tilde{\eta}_i + B_i \tilde{K}_{\eta i}(t) v$. By Lemma 2, $\tilde{S}(t)$ decays to zero exponentially at least as fast as $e^{-\mu_1 \alpha_H t}$ where $\alpha_H = \min(\Re(\sigma(H)))$, and $\tilde{\eta}_i$ decays to zero exponentially for sufficiently large μ_1 and μ_2 . Since $\tilde{K}_{\eta i}(t) = (\mathcal{U}_i(t) - U_i^*) - K_{xi}(\mathcal{X}_i(t) - X_i^*)$, $\lim_{t \rightarrow \infty} \tilde{K}_{\eta i}(t) = 0$ exponentially at least as fast as $\tilde{S}(t)$ for sufficiently large μ_3 by Lemma 4. Thus, $\tilde{K}_{\eta i}(t) v$ decays to zero exponentially, and hence, $\lim_{t \rightarrow \infty} f_i(t) = 0$. Moreover, since $(A_i + B_i K_{xi})$ is Hurwitz, by Lemma 1, for any initial condition $\tilde{x}_i(0)$, $\lim_{t \rightarrow \infty} \tilde{x}_i(t) = 0$ exponentially. Thus, $\lim_{t \rightarrow \infty} \tilde{u}_i(t) = 0$ by (31), and hence $\lim_{t \rightarrow \infty} e_i(t) = 0$ by (30). $\square$

Remark 4. From the proof of Lemmas 2–4, if none of the eigenvalues of S has positive real parts, then Theorem 1 holds for any positive μ_1, μ_2 and μ_3 . Otherwise, we can choose $\mu_2 > \frac{\gamma}{\alpha_H}$ and $\mu_1 > \frac{\gamma}{\alpha_H}$ where $\gamma = \max(\Re(\sigma(S)))$, and $\mu_3 \geq \frac{\mu_1 \alpha_H}{\min(\sigma(Q_i^T Q_i))} > \frac{\gamma}{\min(\sigma(Q_i^T Q_i))}$ for $i = 1, \dots, N$.

Next, we will further consider solving Problem 1 by measurement output feedback control. Let K_{xi} and $K_{\eta i}(t)$ be defined as in the state feedback control law. Since (C_{mi}, A_i) is detectable, let L_i be such that $A_i + L_i C_{mi}$ is Hurwitz. For $i = 1, \dots, N$, we design the following measurement output feedback controller

$$u_i = K_{xi} \xi_i + K_{\eta i}(t) \eta_i \quad (33a)$$

$$\begin{aligned} \dot{\xi}_i &= A_i \xi_i + B_i u_i + E_i \eta_i \\ &\quad + L_i (C_{mi} \xi_i + D_{mi} u_i + F_{mi} \eta_i - y_{mi}). \end{aligned} \quad (33b)$$

We have the following result.

Theorem 2. Given systems (1), (2) and a graph $\bar{\mathcal{G}}$, under Assumptions 1–4, by choosing μ_1, μ_2 and μ_3 sufficiently large, Problem 1 is solvable by the control law composed of (5), (22), and (33).

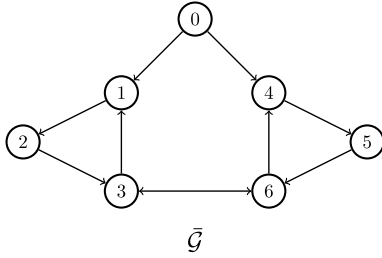


Fig. 1. Communication graph $\bar{G}$.

Proof. Let $\hat{x}_i = \xi_i - x_i$. Then we have

$$\begin{aligned}\dot{\hat{x}}_i &= A_i \xi_i + B_i u_i + E_i \eta_i - A_i x_i - B_i u_i - E_i v + L_i \\ &\quad \cdot (C_{mi} \xi_i + D_{mi} u_i + F_{mi} \eta_i - C_{mi} x_i - D_{mi} u_i - F_{mi} v) \\ &= A_i \hat{x}_i + E_i \tilde{\eta}_i + L_i C_{mi} \hat{x}_i + L_i F_{mi} \tilde{\eta}_i \\ &= (A_i + L_i C_{mi}) \hat{x}_i + (E_i + L_i F_{mi}) \tilde{\eta}_i.\end{aligned}\quad (34)$$

Since $\lim_{t \rightarrow \infty} \tilde{\eta}_i(t) = 0$ exponentially and $(A_i + L_i C_{mi})$ is Hurwitz, by Lemma 1, $\lim_{t \rightarrow \infty} \hat{x}_i(t) = 0$ exponentially. Note that in this case, (29) and (30) still hold. Next, similarly to (31), a simple calculation gives

$$\begin{aligned}\tilde{u}_i &= K_{xi}(\tilde{x}_i + \hat{x}_i + X_i^* v) + K_{\eta i}(t)(\tilde{\eta}_i + v) - U_i^* v \\ &= K_{xi} \tilde{x}_i + K_{xi} \hat{x}_i + K_{\eta i}(t) \tilde{\eta}_i + \tilde{K}_{\eta i}(t) v.\end{aligned}\quad (35)$$

The remaining proof follows the proof of Theorem 1 by noticing that $K_{xi} \hat{x}_i$ will also decay to zero exponentially. $\square$

5. An example

Given a leader system of the following form

$$\dot{v} = \begin{pmatrix} 0 & \theta \\ 0 & 0 \end{pmatrix} v, \quad v(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (36)$$

which produces a ramp function with the slope θ , let us consider the position tracking problem of multiple robotic systems with $N = 6$ whose motion equations are given by Cheah, Hou, and Slotine (2009)

$$\dot{x}_i = \begin{pmatrix} 0 & 1 \\ 0 & -\zeta_i \end{pmatrix} x_i + \begin{pmatrix} 0 \\ b_i \end{pmatrix} u_i \quad (37a)$$

$$e_i = x_{i1} - y_0 \quad (37b)$$

$$y_{mi} = \begin{pmatrix} 1 & 0 \end{pmatrix} x_i \quad (37c)$$

$$i = 1, 2, \dots, 6$$

where $x_i = \text{col}(x_{i1}, x_{i2}) \in R^2$ with x_{i1} and x_{i2} representing the position and velocity variables, respectively, $u_i \in R$ is the control input, and $y_0(t) = [1, 0]v = \theta t$ is the reference position determined by the leader system. ζ_i and b_i denote the damping and mass coefficients, respectively. Let $\zeta_i = 0.1 + i \cdot 0.2$ and $b_i = 1$. The communication graph is shown in Fig. 1. It can be easily verified Assumptions 1–4 are satisfied. Let $\theta = 5$ m/s. Both state feedback and measurement output feedback control laws are designed. The control gains for the distributed observer (5) are chosen as $\mu_1 = \mu_2 = 10$. The control gain for the online regulator equation solver (22) is chosen to be $\mu_3 = 40$. The feedforward gain of (28) and (33a) is set to be $K_{xi} = [-10, -5]$. The observer gain of (33b) is selected to be $L_i = \text{col}(-2, -3)$. The initial values are given as, for $i = 1, \dots, 6$, $x_{i1}(0) = 2 + 2 \cdot i$, $x_{i2}(0) = 0$, $S_i(0) = 0$, $\eta_i(0) = 0$, $\zeta_i(0) = 0$, $\xi_i(0) = 0$. The position tracking errors are shown in Figs. 2 and 3, where $\hat{e}_i$ s represent the tracking errors under the non-

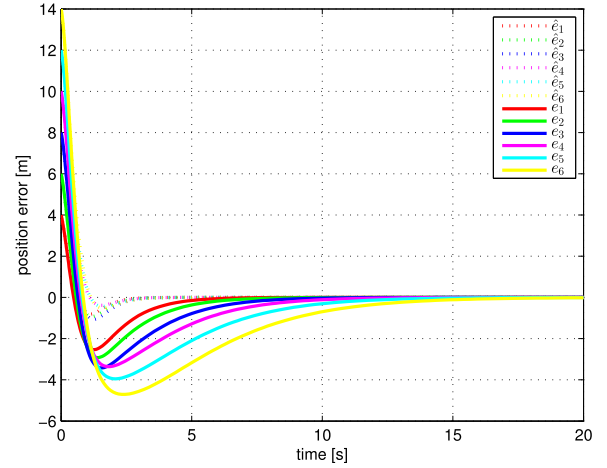


Fig. 2. Position tracking errors by state feedback control.

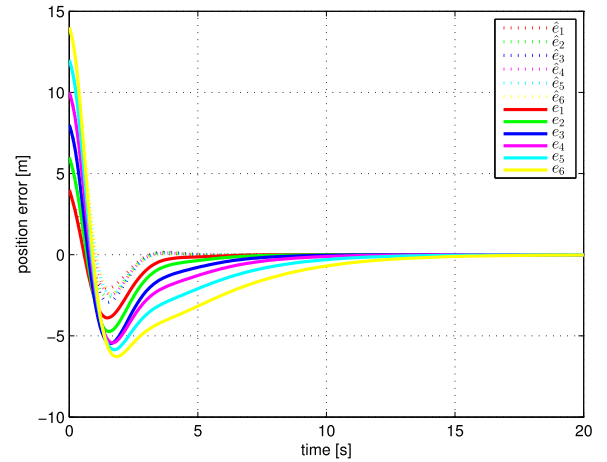


Fig. 3. Position tracking errors by measurement output feedback control.

adaptive controllers when S is known and e_i s represent the tracking errors under the adaptive controllers when S is estimated. It can be seen that the tracking performance using the adaptive controller is almost as good as that of using the non-adaptive controller.

6. Conclusion

A drawback for the existing approach to dealing with the cooperative output regulation problem of linear multi-agent systems is that the result relies on the assumption that the system matrix of the leader system is available to each follower. We have removed this assumption in this paper by employing an adaptive distributed observer to estimate the system matrix of the leader system adaptively. Moreover, we have developed an adaptive algorithm to calculate the solution of the regulator equations without knowing the exact knowledge of the system matrix of the leader system. Based on the adaptive distributed observer and the adaptive solution of the regulator equations, we have synthesized both adaptive state feedback and adaptive measurement output feedback controllers to solve the cooperative output regulation problem.

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